MICROMAX SUGAR MILLS ERP · LABORATORY GUIDE

Laboratory & production

Setting the laboratory up, the analyses through the crushing day, vehicle core samples, mill stoppages, and the daily manufacturing report that turns the day's cane into sugar, by-products and recovery.

1 · Setup
Streams, limits, settings, norms.

2 · Hourly
Brix, pol, pH; purity worked out.

3 · Vehicles
Core samples → CCS, recovery.

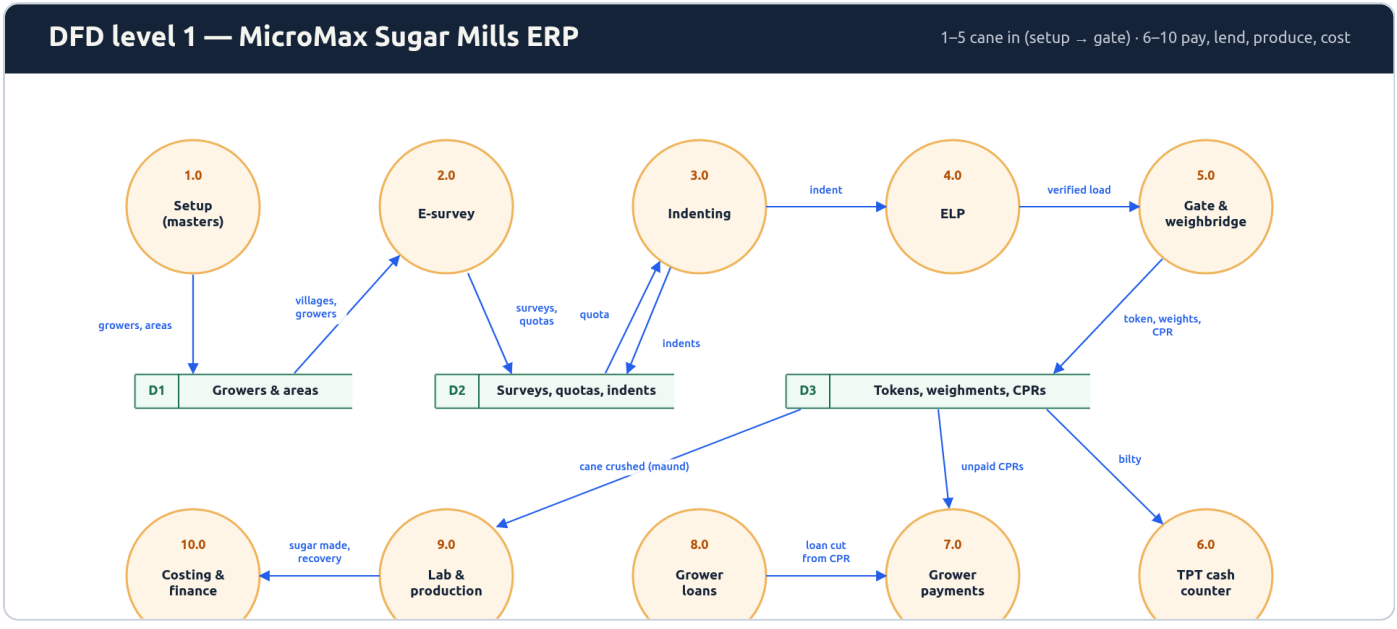
4 · Stoppages
Category, section, reason.

5 · DMR
Prepare, approve, post, print.

Who: **Lab Chemist** (samples, stoppages, DMR), **Production Manager** (approve, post production), **Cane Admin** (setup).

DFD level 1 — MicroMax Sugar Mills ERP

1–5 cane in (setup → gate) · 6–10 pay, lend, produce, cost

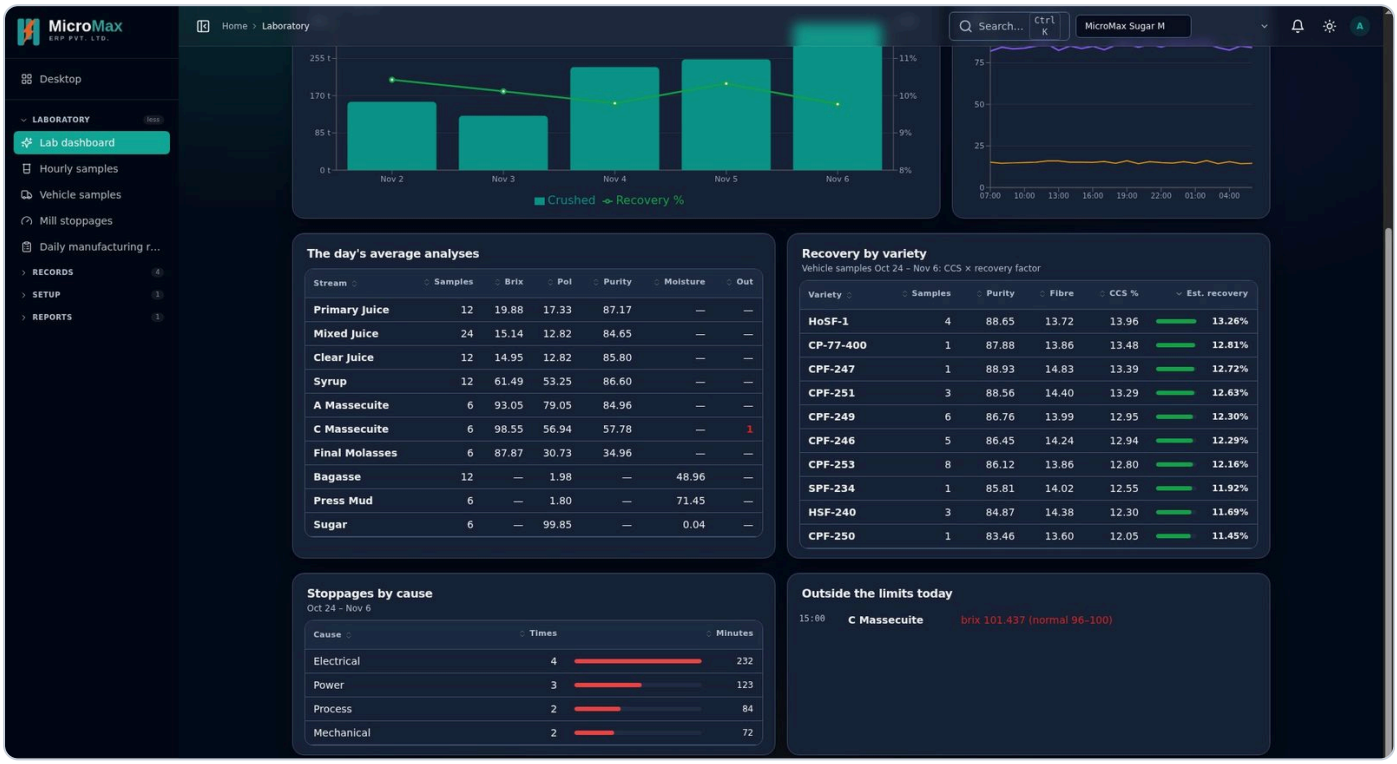


DFD level 1 of the whole ERP — lab & production (9.0) and costing (10.0) at the end of the cane flow.

1 · Laboratory setup

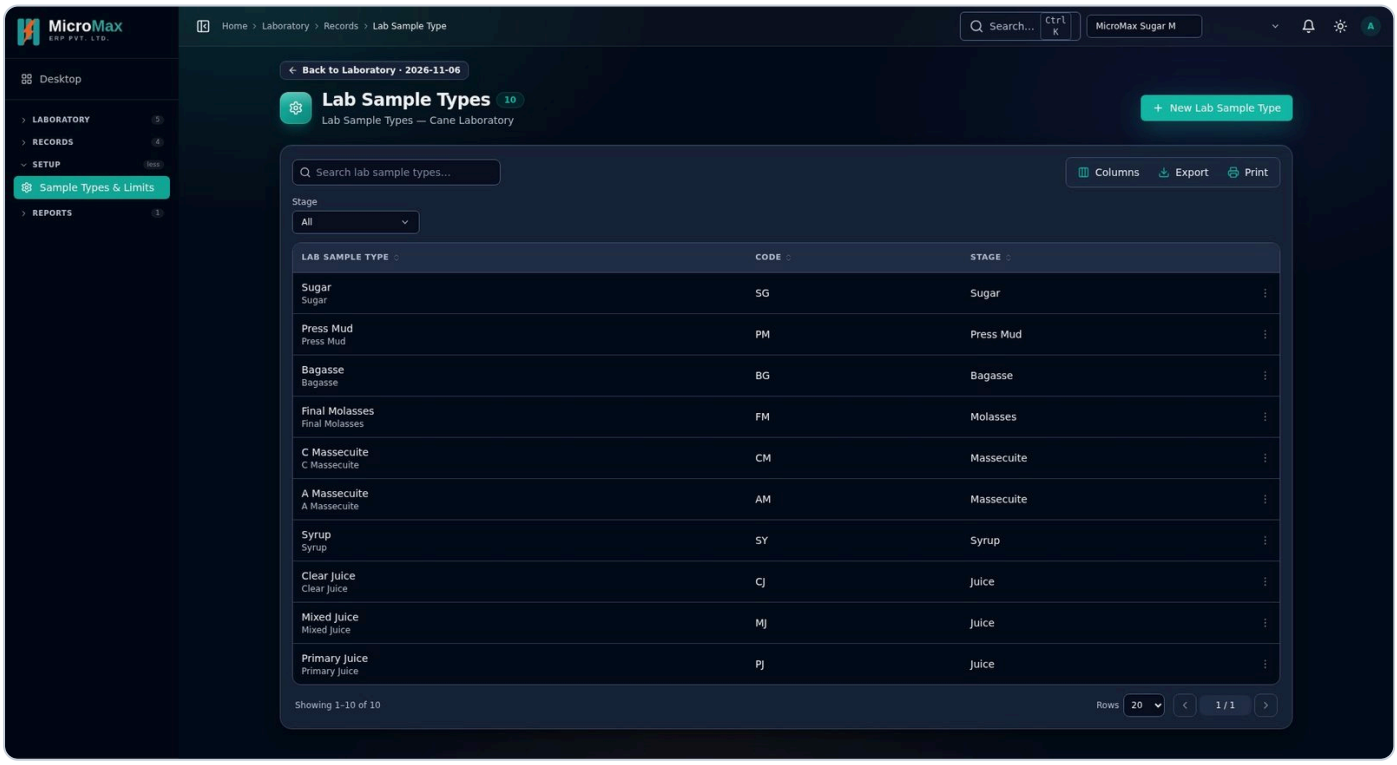
1 Lab dashboard

Crushed and recovery by day, the day's averages per stream, recovery by variety, stoppages by cause and samples outside the limits.



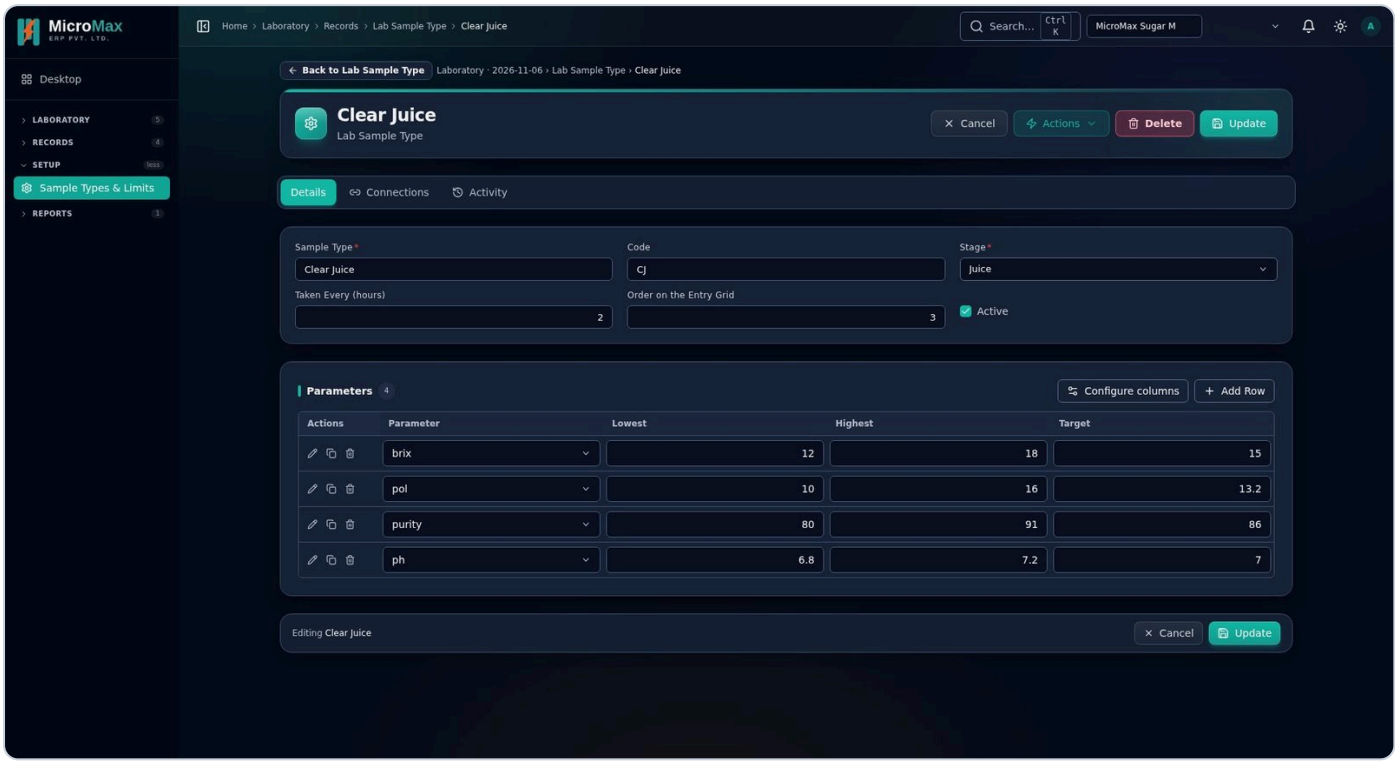
2 Sample types

Laboratory → Setup → Sample Types & Limits: ten streams from primary juice to sugar, each with its stage and how often it is taken (every 1, 2 or 4 hours).



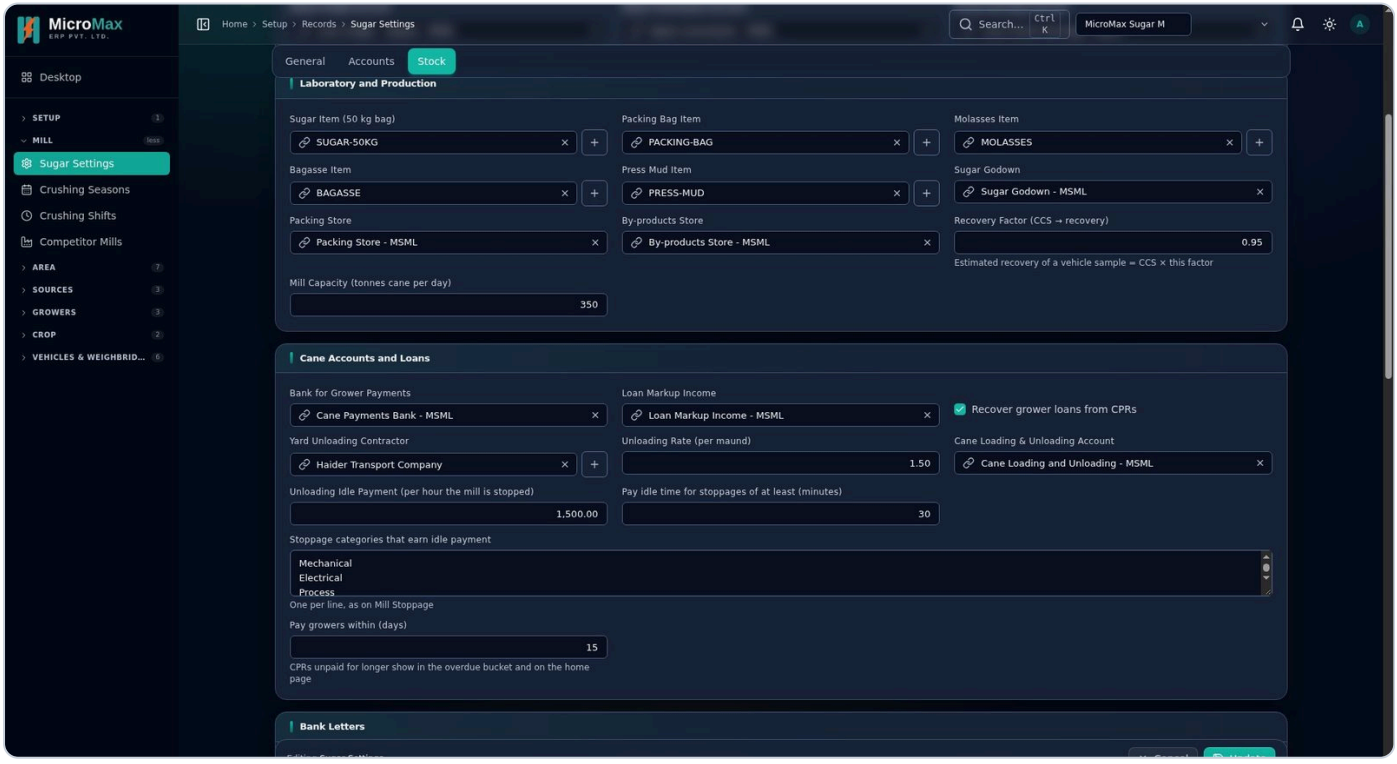
3 Limits per stream

Each parameter's lowest, highest and target. Clear juice: brix 12–18, pol 10–16, purity 80–91, pH 6.8–7.2. A value outside is flagged red.



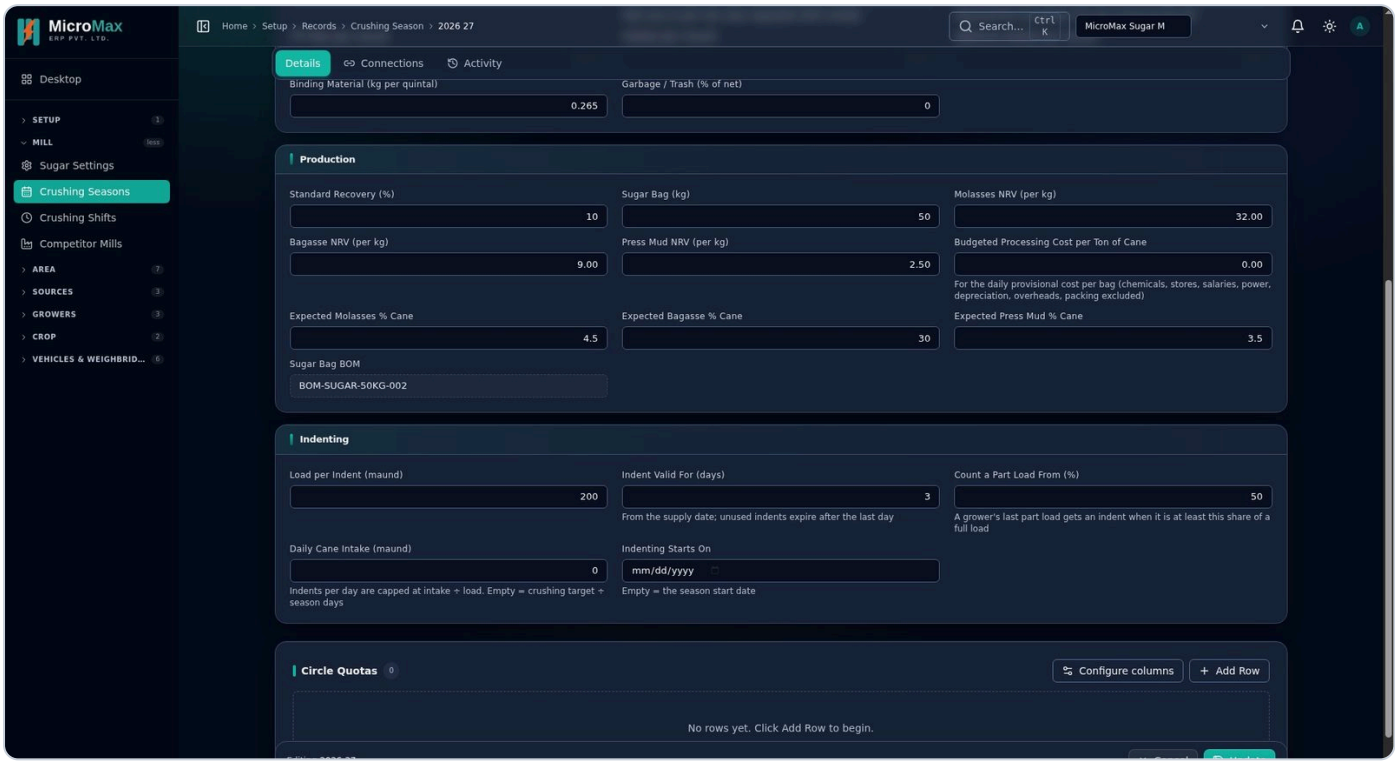
4 Sugar Settings — Laboratory and Production

The sugar, bagasse, molasses, press mud and packing items, the sugar and by-product godowns, the recovery factor and the mill capacity (t/day).



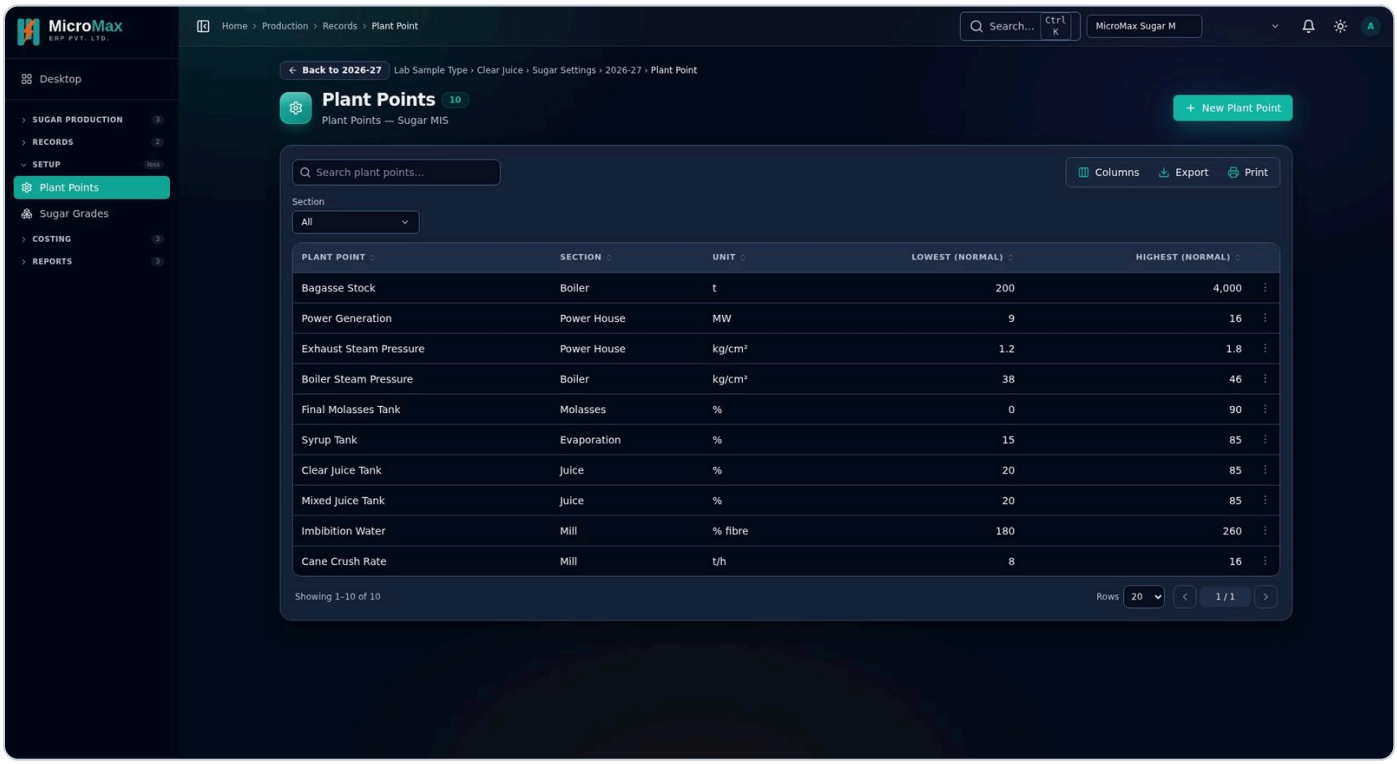
5 Season production norms

Crushing Season → Production: standard recovery, bag size, expected molasses / bagasse / press mud % cane and by-product NRV.



6 Plant points

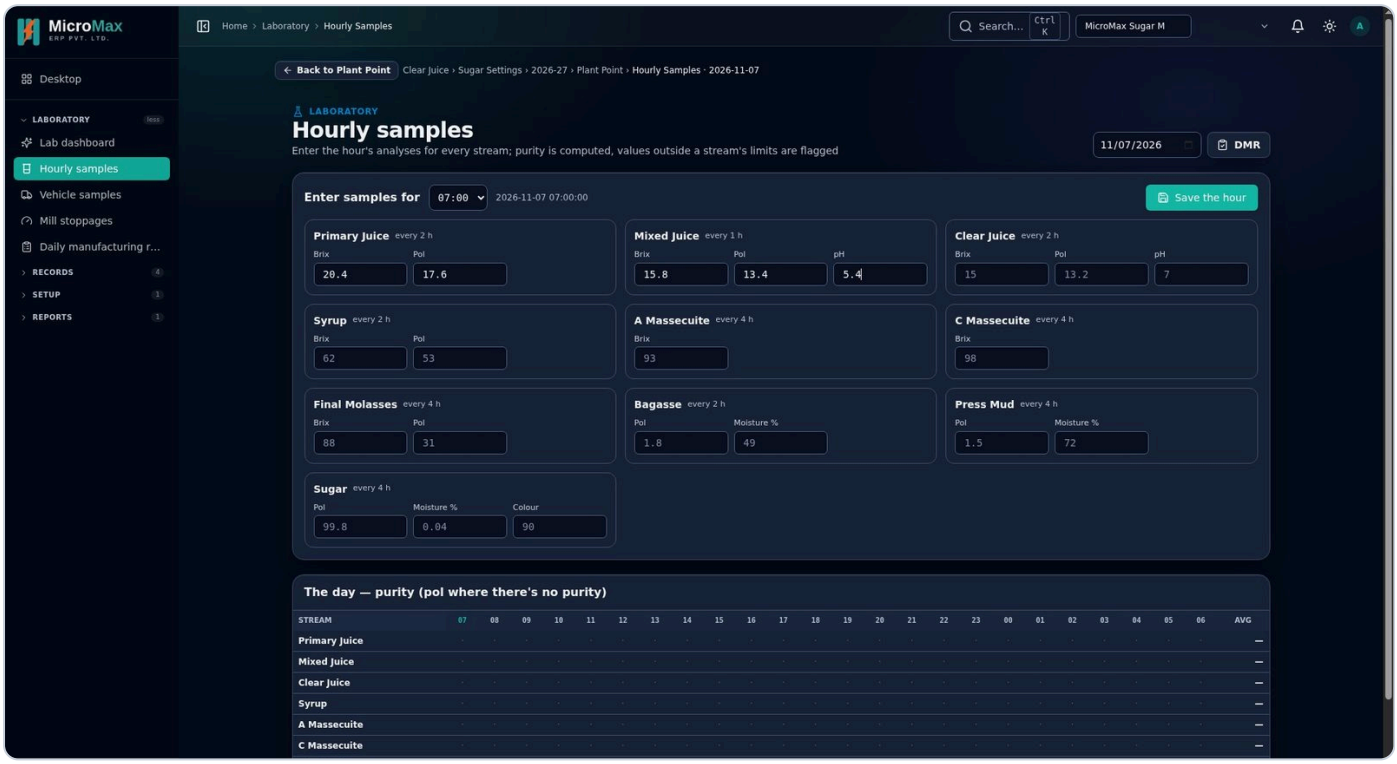
Production → Setup → Plant Points: tanks, boilers, power and crush rate with their normal range.



2 · The crushing day (07:00 → 07:00)

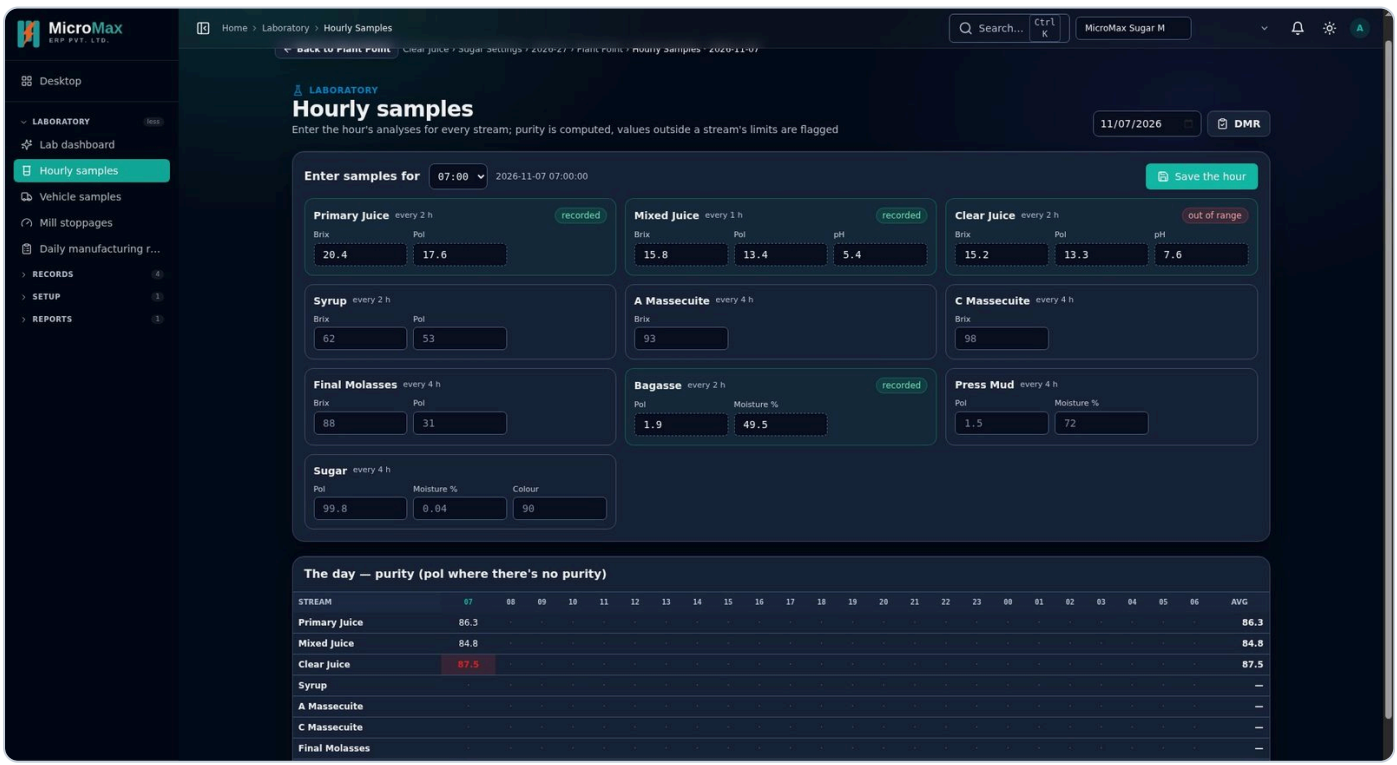
7 Hourly samples

Laboratory → Hourly samples. Pick the hour; the streams due that hour are bright. Enter brix, pol, pH, moisture as tested — purity is worked out.



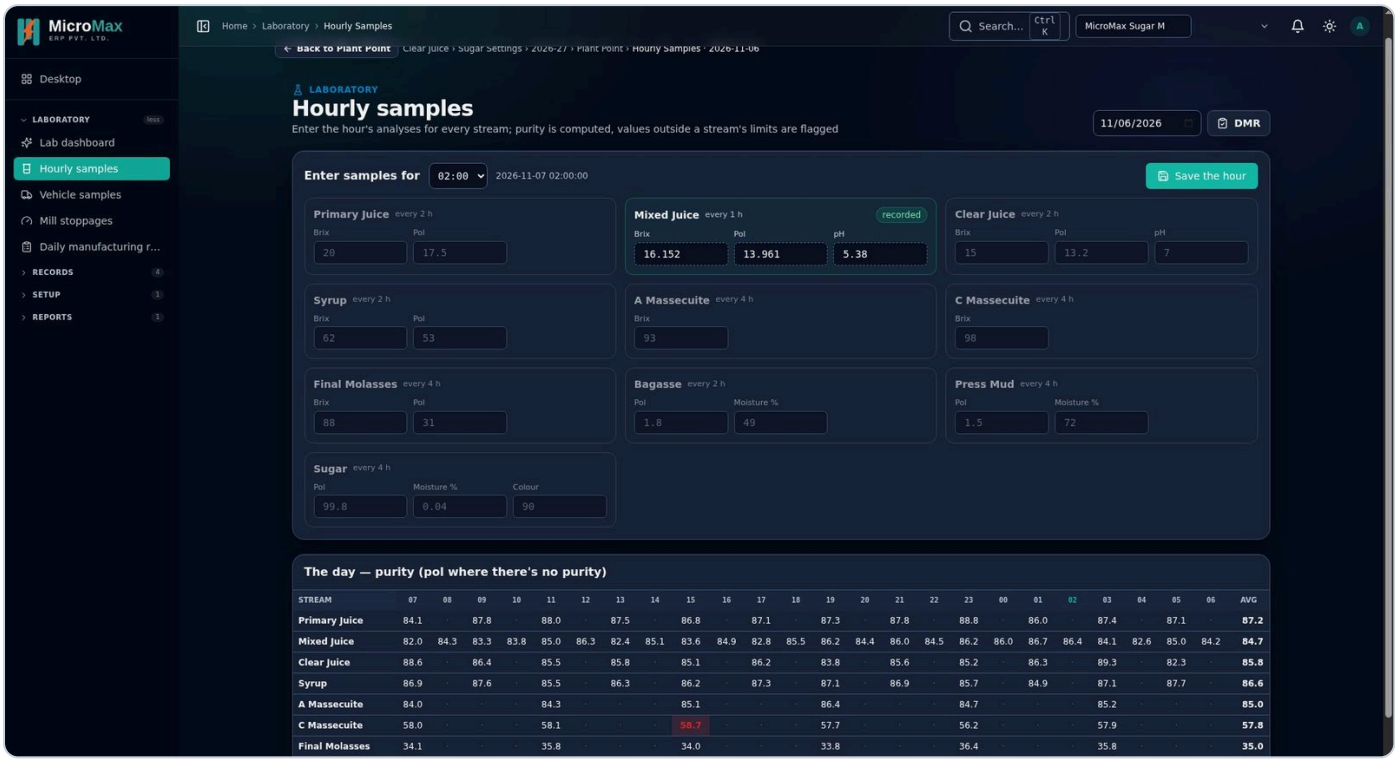
8 Save the hour

Save the hour. A value outside its limits is saved but marked **out of range** (here clear juice pH 7.6) and shows red on the day grid and the MIS.



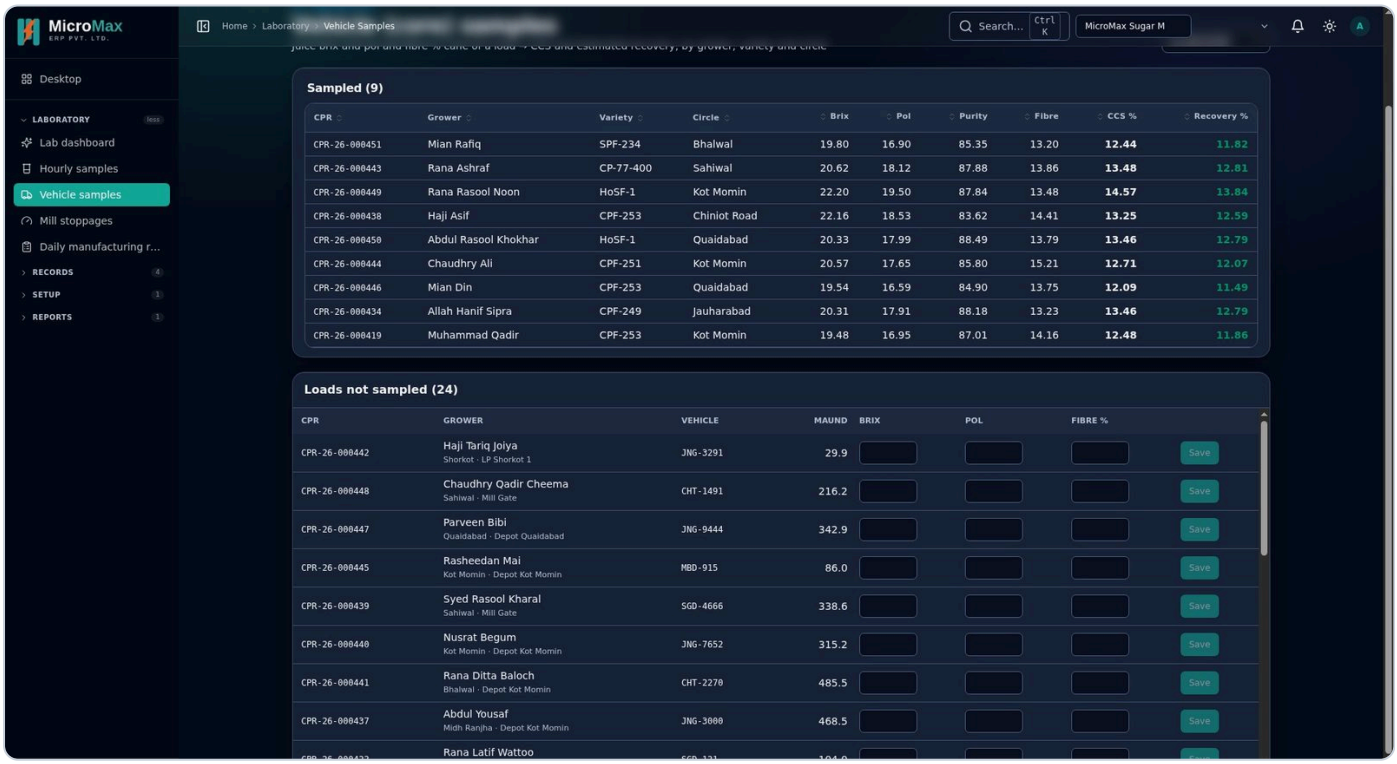
9 The whole day

The grid of 24 hours × 10 streams; click any cell to open that hour.



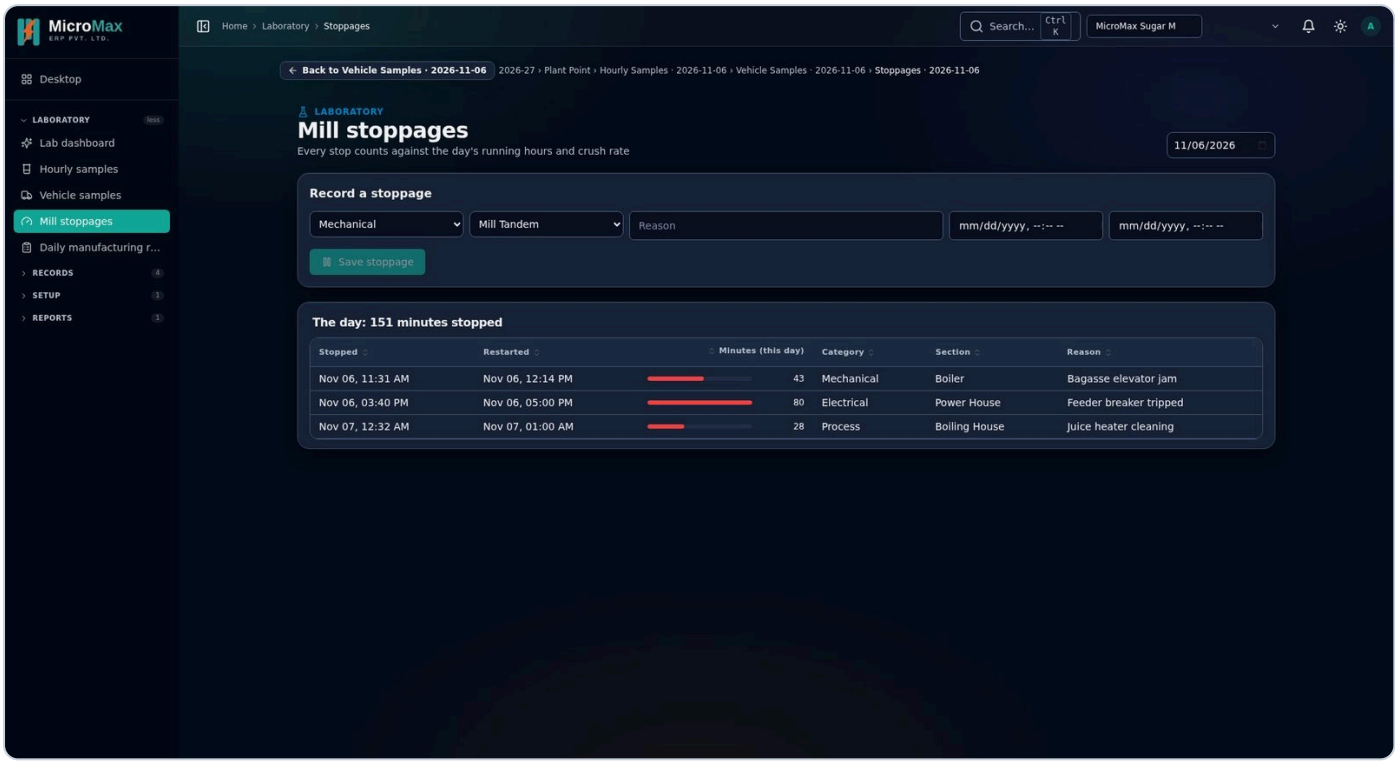
10 Vehicle core samples

Laboratory → Vehicle samples: loads not yet sampled with their CPR. Enter brix, pol and fibre and Save — CCS % and the estimated recovery are worked out.



11 Mill stoppages

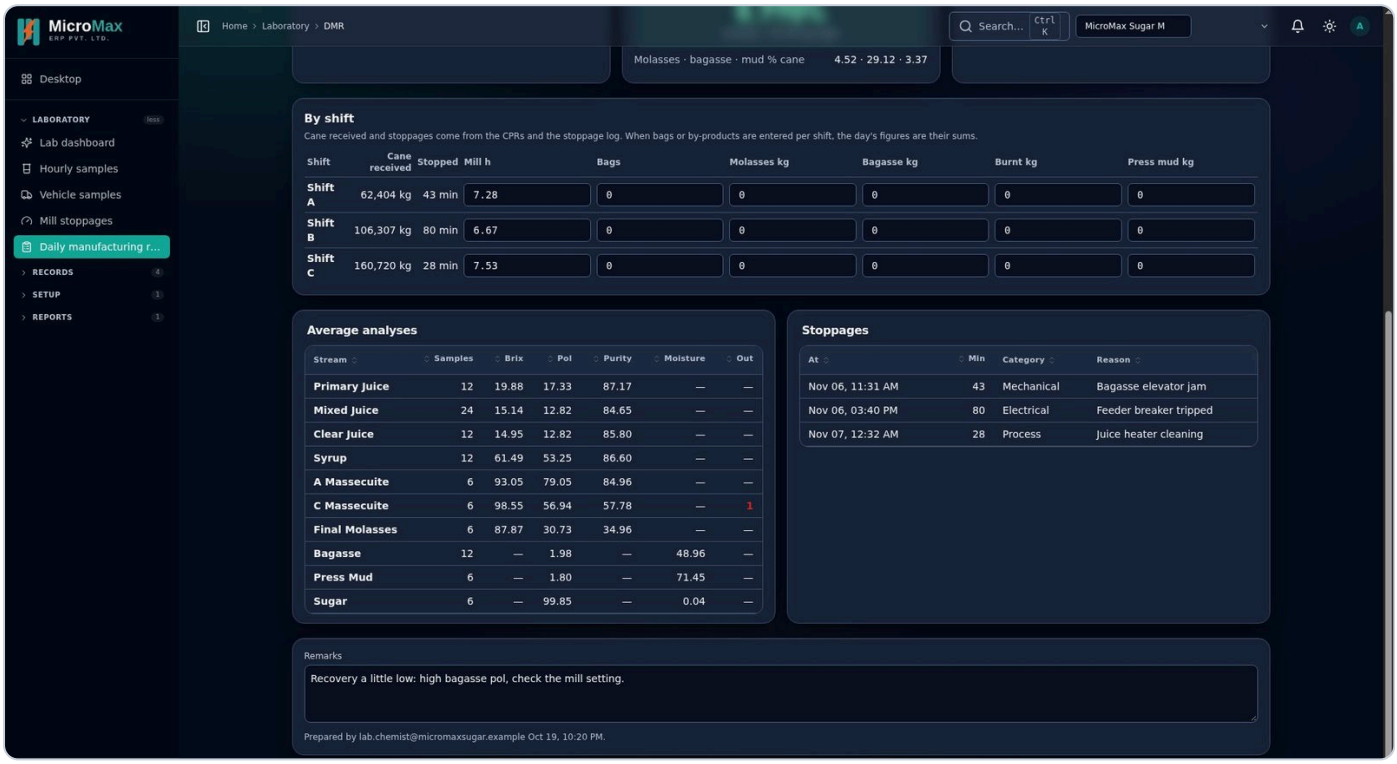
Record each stop: category, section, reason, stopped and restarted. Open stops show a red banner with **Restarted now**.



3 · Daily manufacturing report

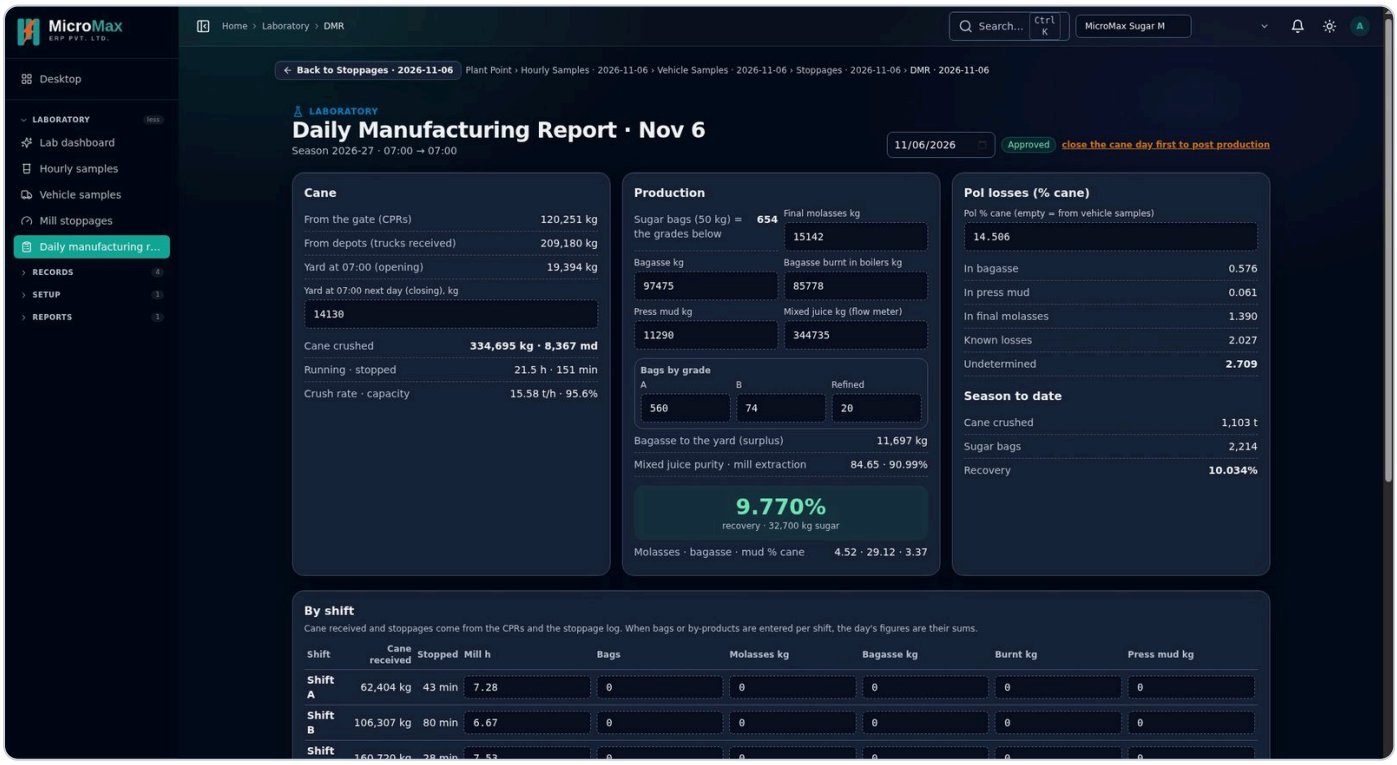
12 Prepare the DMR

After 07:00, Laboratory → Daily manufacturing report. Cane comes from the CPRs; enter the yard closing, bags by grade and by-products, by shift. Save, then **Prepare**.



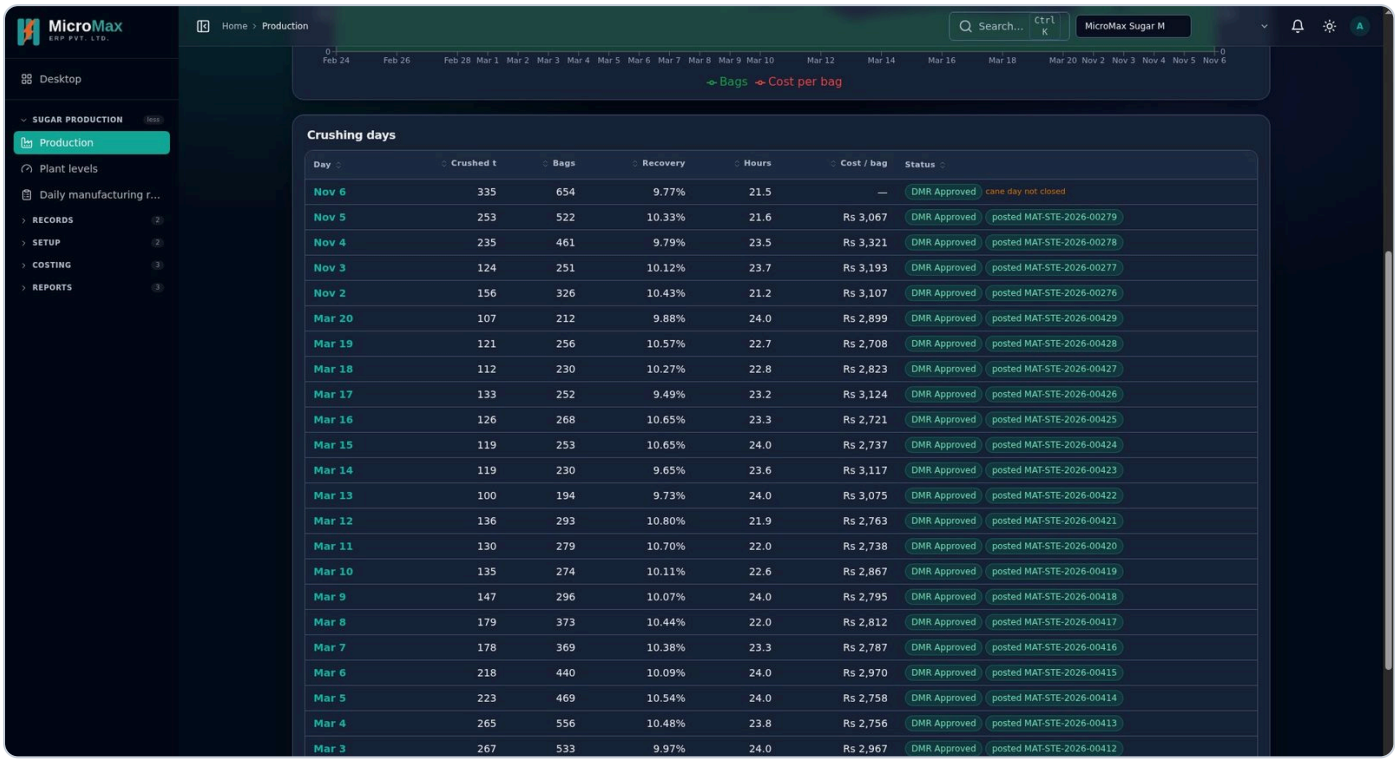
13 Approve

The Production Manager checks and **Approves**. When the cane day is closed, **Post production** makes the stock entries.



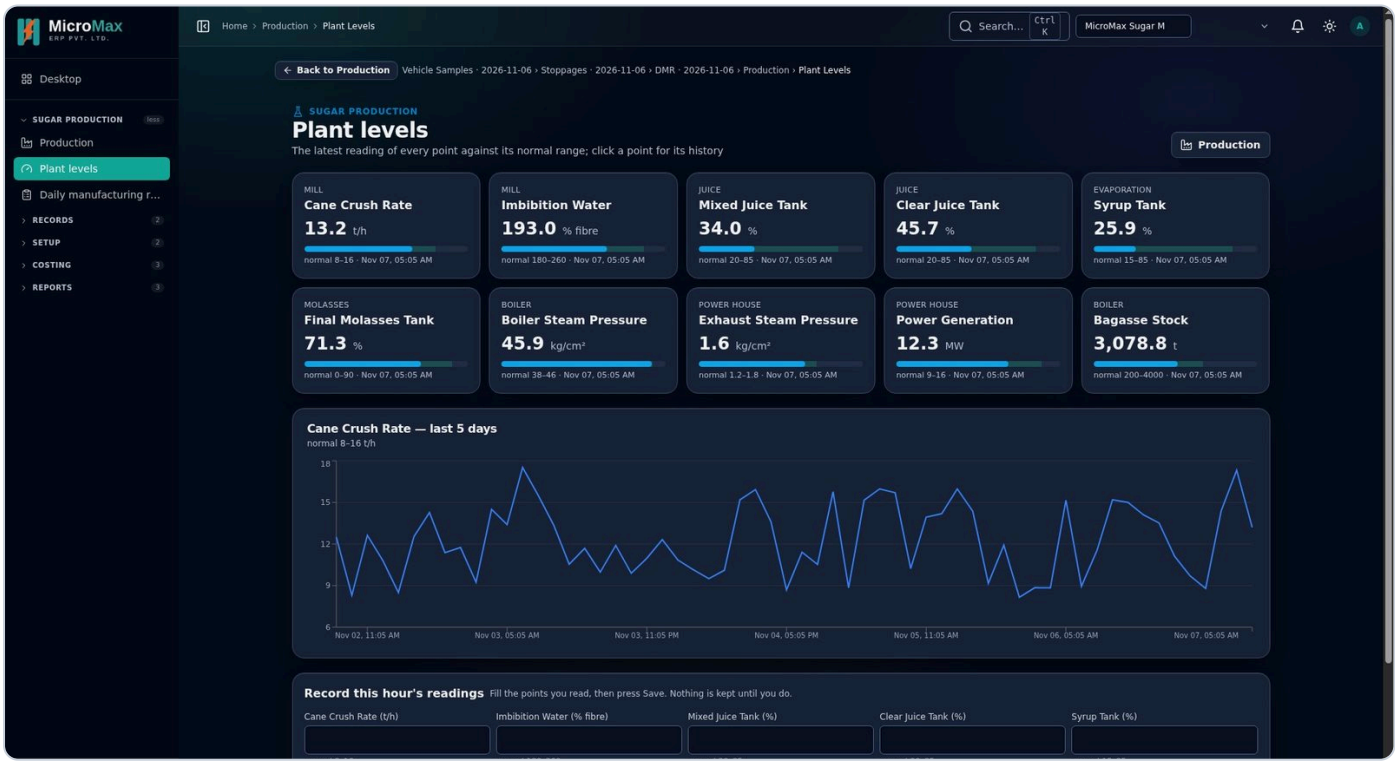
14 Production

Production: each day's crushed cane, bags, recovery, hours, cost per bag and the posted stock entry.



15 Plant levels

The latest reading of every plant point against its normal range, with the history.



The DMR sheet



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DAILY MANUFACTURING REPORT
Crushing day 06 Nov 2026 (07:00 → 07:00) · season 2026-27 · DMR-2026-11-06 · **Approved**

335 t	654	9.77 %	15.58 t/h	21.5 h	95.6 %
CANE CRUSHED	SUGAR BAGS (50 KG)	RECOVERY	CRUSH RATE	RUNNING · 151 MIN STOPPED	CAPACITY USED

Cane

From the gate (CPRs)	120,251 kg
From depots	209,180 kg
Received	329,431 kg
Yard opening	19,394 kg
Yard closing	14,130 kg
Crushed	**334,695 kg 8,367 md**

Production

Sugar A	560 bags
Sugar B	74 bags
Sugar Refined	20 bags
Sugar	**654 bags · 32,700 kg**
Final molasses	15,142 kg · 4.52 %
Bagasse	97,475 kg · 29.12 %
· burnt / to yard	85,778 / 11,697 kg
Press mud	11,290 kg · 3.37 %

Performance (% cane)

Pol % cane	14.506
Mixed juice purity	84.65
Mill extraction	90.99
Pol lost in bagasse	0.576
· in press mud	0.061
· in final molasses	1.390
· undetermined	2.709
Total losses	**4.736**

By shift

Shift	Cane received kg	Stopped min	Mill h	Bags	Molasses kg	Bagasse kg	Burnt kg	Press mud kg
Shift A	62,404	43	7.28					
Shift B	106,307	80	6.67					
Shift C	160,720	28	7.53					

Laboratory averages

Stream	Samples	Brix	Pol	Purity	Moisture	Out
Primary Juice	12	19.88	17.33	87.17		
Mixed Juice	24	15.14	12.81	84.65		
Clear Juice	12	14.95	12.82	85.80		
Syrup	12	61.49	53.25	86.60		
A Massecuite	6	93.05	79.05	84.96		
C Massecuite	6	98.55	56.94	57.78		1
Final Molasses	6	87.87	30.73	34.96		
Bagasse	12		1.98		48.96	
Press Mud	6		1.80		71.45	
Sugar	6		99.85		0.04	

Stoppages

Category	Min
Electrical	80
Mechanical	43
Process	28
Total	**151**

Season to date

Cane crushed	1,103 t
Sugar bags	2,214
Recovery	10.034 %
Running hours	111.5

Remarks: Recovery a little low: high bagasse pol, check the mill setting.

Lab Chemist — Ayesha Siddiqa
Production Manager — Administrator
General Manager

MicroMax Sugar Mills Ltd. · Daily manufacturing report DMR-2026-11-06 · Printed 02 Nov 2026 02:20 by Administrator

Print DMR on the day's report (or Print in the desk) gives one A4 sheet with the QR code back to the day.

Cane: from the gate and depots, yard opening and closing, crushed (kg and maunds).

Production: sugar bags by grade, molasses, bagasse (burnt and to the yard), press mud — each as % cane.

Performance: pol % cane, mixed juice purity, mill extraction, pol losses and the undetermined loss.

By shift, **lab averages** with samples out of range, **stoppages** by category and **season to date**.

Signed by the Lab Chemist, the Production Manager and the General Manager.

If something goes wrong

You see	Do this
A value shows red / *out of range*	Check the analysis; it is saved, but appears on the DMR and the MIS.
“Close the cane day first to post production”	Cane Procurement → Day close for that day (earlier days first).
Prepare refused	Enter the yard closing and the sugar bags, then Save.
A stoppage still open	Laboratory → Mill stoppages → Restarted now.
Recovery looks wrong	Check the bags by grade and the yard closing; recovery = sugar ÷ cane crushed.

Screens from the MicroMax test system with demo data (2 November 2026).